

A Zero-Parameter EEG-fNIRS Rule Ensemble for Cross-Subject Emotion Tracking

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Abstract

MER2026 Track 4 evaluates second-level valence and arousal prediction from synchronized EEG and fNIRS recordings in a cross-subject setting. We present V130, a zero-parameter ensemble that combines resting-state trait votes, task-state temporal descriptors, and confidence-aware decision rules. The training partition contains 24 subjects and 360 subject-video sequences, so the system emphasizes stable directional evidence and physiologically motivated feature choices. Across more than 400 development trials, subject-disjoint evaluation favored the rule ensemble over learned feature selection for cross-subject transfer. V130 reduced public Overall MAE from 28.37 to 26.12. On the final set, it obtained Valence MAE 24.62, Arousal MAE 26.41, and Overall MAE 25.52, ranking seventh. Variance analysis indicates that stable resting-state features explain 64-94% of cross-subject variation, while task-state descriptors primarily capture within-video variation. These results indicate that physiology-guided rule ensembles are a practical option for small-cohort cross-subject emotion recognition.

CCS Concepts

• **Computing methodologies:** Machine learning; Human-centered computing; Human-computer interaction;

Keywords

physiological emotion recognition, EEG-fNIRS, cross-subject generalization, rule ensemble, MER-PS

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1 INTRODUCTION

Physiological emotion recognition complements observable cues such as speech and facial expression with direct measurements of neural and hemodynamic activity. EEG offers millisecond temporal resolution, while fNIRS tracks slower oxygenation changes and can provide complementary spatial evidence [2]. MER2026 Track

4 (MER-PS) brings these modalities together in a dynamic cross-subject benchmark [1]. The task is difficult not because labeled seconds are scarce, but because independent subjects are scarce and physiological baselines differ sharply across people.

Our initial course project explored EEGNet-style networks, pre-trained brain encoders, spiking models, and large-model adapters. Those experiments provided useful comparisons, while subject-disjoint evaluation favored lower-capacity physiological rules. We therefore reformulated the problem around a narrower question: which low-capacity physiological decisions remain stable when the test subjects are entirely unseen? The resulting system learns no weights. It combines resting-state trait indicators, robust video-level state descriptors, and asymmetric decision rules.

The contributions are threefold. First, we introduce a zero-parameter, three-stage ensemble that produces second-level valence and arousal trajectories. Second, we show that data-selected features can improve local MAE while degrading the public leaderboard, whereas physiology-guided features transfer better. Third, the final system achieves 25.52 Overall MAE (24.62 valence and 26.41 arousal), corresponding to seventh place.

1.1 Task and evaluation

The full MER-PS collection contains synchronized recordings from 30 participants watching 15 emotion-eliciting clips. EEG has 64 channels sampled at 1000 Hz and fNIRS has 51 channels sampled at 47.62 Hz. Participants continuously report valence and arousal with a joystick at 1 Hz [1]. The challenge training data expose 24 participants; the remaining subjects are used for held-out evaluation. Predictions lie on the 1-255 annotation scale. We report mean absolute error for valence (V-MAE) and arousal (A-MAE), with Overall MAE equal to their average.

We treat each subject-video trajectory, rather than each second, as an independent statistical unit. This distinction matters: the 24 x 15 design yields only 360 sequences for model selection, while the engineered feature pool exceeds 626 candidates. Repeated seconds increase sample count without eliminating subject dependence, creating a high risk of leakage and unstable feature ranks.

1.2 Related work

Physiological affect recognition has traditionally been evaluated on datasets such as DEAP, where EEG and peripheral signals are mapped to categorical or dimensional emotion labels [7]. Modern cross-subject methods learn electrode graphs, domain-invariant embeddings, or subject-specific normalizers; self-organized graph models, for example, explicitly adapt connectivity structure to reduce subject mismatch [4]. Domain-adversarial training offers a general mechanism for suppressing domain identity [10], but it

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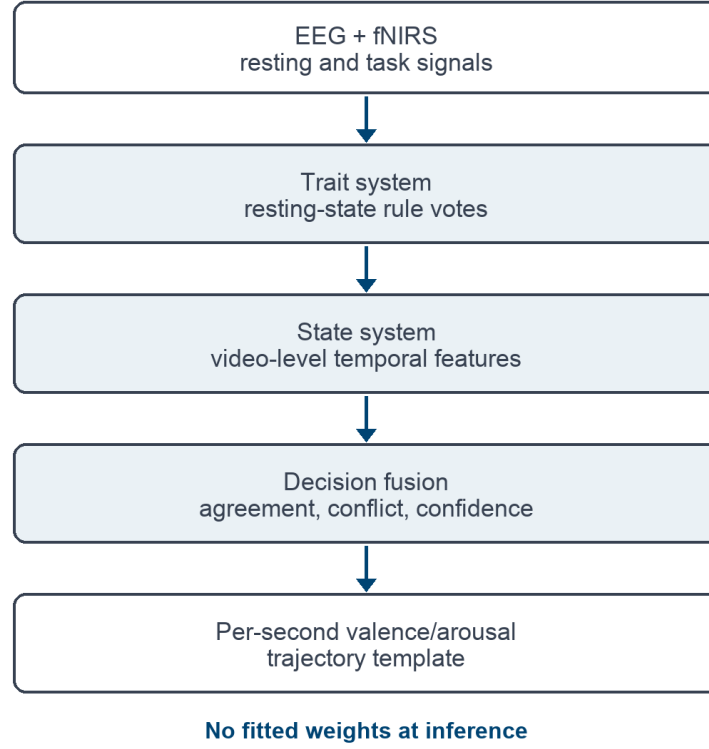


Figure 1: Three-stage inference pipeline. Trait and state subsystems emit directional votes; deterministic fusion converts the votes into per-video templates and second-level trajectories.

requires enough independent domains to distinguish invariant signal from noise. With only 24 training subjects, that assumption is fragile.

Concurrent EEG-fNIRS systems add a second source of variation. EEG responds rapidly to electrical activity, whereas fNIRS reflects a delayed hemodynamic process; alignment, motion artifacts, probe coverage, and baseline oxygenation therefore affect fusion [2]. REFED provides synchronized EEG-fNIRS recordings with real-time labels and illustrates the value of multimodal dynamic benchmarks [6]. Learned encoders such as EEGNet [9] and spiking architectures [8] can model local temporal structure, but model capacity alone does not guarantee transfer when subject identity dominates the embedding.

Our contribution is complementary to these representation-learning approaches. Instead of estimating a subject-invariant latent space from a small cohort, we constrain inference to directional votes and video templates whose decisions can be inspected directly. This trades within-subject resolution for lower selection variance. The challenge lineage has progressively expanded multimodal emotion recognition from robust classification to open-vocabulary and physiological understanding [11, 12]; MER-PS makes cross-subject calibration a central part of that progression.

2 ZERO-PARAMETER SYSTEM

Figure 1 summarizes the deployed pipeline. All thresholds and mappings are fixed before final-set evaluation. The system first estimates stable subject tendencies from resting EEG-fNIRS, then derives coarse task-state evidence for each video, and finally reconciles the two through deterministic rules. The output is a per-video trajectory template at the required 1 Hz rate.

2.1 Signal construction

EEG recordings are divided into one-second epochs. After a Hanning window and fast Fourier transform, we aggregate log power in delta (0.5-4 Hz), theta (4-8 Hz), alpha (8-13 Hz), beta (13-30 Hz), and gamma (30-45 Hz) bands for each channel. We additionally compute frontal asymmetry and band-power ratios used by the rule bank. For fNIRS, one-second windows are summarized by hemoglobin-type and channel statistics, including mean level, dispersion, slope, and late-window persistence. All comparisons are made after robust within-subject centering so absolute amplitude differences do not become direct subject identifiers.

The system deliberately separates three time scales. Resting summaries represent trait-like subject baselines. Whole-video descriptors represent the state response to a particular stimulus. Per-second values are generated only after stable directions are chosen; we do not fit an independent high-dimensional regressor at every

Table 1: Variance decomposition motivates trait-first prediction.

Feature family	Across subjects	Within video	Transfer
Trait (rest)	64-94%	6-36%	Stable
State (task)	5-15%	85-95%	Weak

time point. For fNIRS, lagged summaries are considered because hemodynamic responses trail EEG. The final rules use direction and confidence rather than a fitted subject-specific lag coefficient, avoiding another unstable degree of freedom.

2.2 Trait subsystem

The trait subsystem operates on the resting recording that precedes each clip. The initial V84 bank contains 14 directional rules: nine EEG rules built from theta/alpha/beta power ratios and five fNIRS rules built from HbO/HbR baseline variability. Each rule casts a high/low vote for valence or arousal; ties retain the global video prior. Because rules depend on signs, ranks, and within-subject normalization rather than fitted coefficients, a new subject can be processed without fine-tuning.

Across representative features, resting-state descriptors attribute 64-94% of variance to between-subject differences. Task-state descriptors show the opposite pattern: 85-95% of their variance occurs within videos. Trait signals therefore provide a stable subject anchor, while state signals are used only as conditional modifiers.

2.3 State subsystem

For task EEG and fNIRS, we summarize each channel-band trajectory with latency, linear trend, coefficient of variation (CV), habituation, zero-crossing rate, peak time, and late sustain. Exhaustive screening found that late sustain and CV were more repeatable than whole-clip slopes; for one development configuration, late sustain reached Spearman $\rho = 0.510$ with valence. Yet the top-10 features selected on all 24 subjects overlapped by only 33% with those selected on an 18-subject subset. We consequently use state descriptors as coarse video-level direction labels, not as unconstrained second-level regressors.

2.4 Asymmetric fusion and V130

The fusion stage compares trait and state directions. Agreement strengthens the deviation from a video template; disagreement attenuates it; low-confidence decisions revert to a robust fallback. Arousal supports asymmetric amplification because its trait direction accuracy is at least 75%. For a within-subject normalized state value z , the boost is $\min(0.75, 2.0c)$ when trait and state directions agree, where c is rule confidence, and zero otherwise. Valence is not boosted in the same way because its trait direction accuracy is only 45.8%, which would systematically amplify errors.

Operationally, each rule returns a sign s in $\{-1, 0, +1\}$ and a bounded confidence c . Votes are aggregated by median sign, making one extreme channel unable to dominate the decision. If trait and state signs agree, the video prior is shifted in that direction. If they conflict, the prior is kept or mildly attenuated. If too few rules fire, the system switches to the fallback descriptor assigned to that video. Final values are clipped to the 1-255 scale and broadcast

through a smoothed template so isolated feature spikes cannot create implausible annotation jumps.

V130 retains 11 theory-supported EEG rules for six videos and uses nine fallback features for the remaining clips. No regression weight, threshold, or fusion coefficient is optimized on the final test subjects; the deployed inference graph has zero trainable parameters.

3 EXPERIMENTS AND RESULTS

3.1 Development protocol

We ran more than 400 controlled trials spanning compact neural regressors, spiking networks, large-model adapters, hand-engineered features, and rule ensembles. Local analysis used subject-disjoint splits whenever possible. Public CodaBench feedback was reserved for comparing complete system versions rather than tuning individual thresholds. Because several neural runs used different normalization or validation protocols, we treat their losses as diagnostic evidence and do not present them as directly comparable leaderboard scores.

3.2 Main result

The global-median V13 baseline scored 28.37 Overall MAE on the public set. Rule development reduced this to 26.12, an absolute gain of 2.25 points (7.9%). On the final set, V130 achieved 24.62 V-MAE and 26.41 A-MAE, for 25.52 Overall MAE and seventh place. Relative to V13's public score, the final overall error is 10.0% lower. The public-to-final shift increased valence error by 1.12 but decreased arousal error by 2.34, suggesting that the asymmetric arousal rules transferred better than the valence subtype templates.

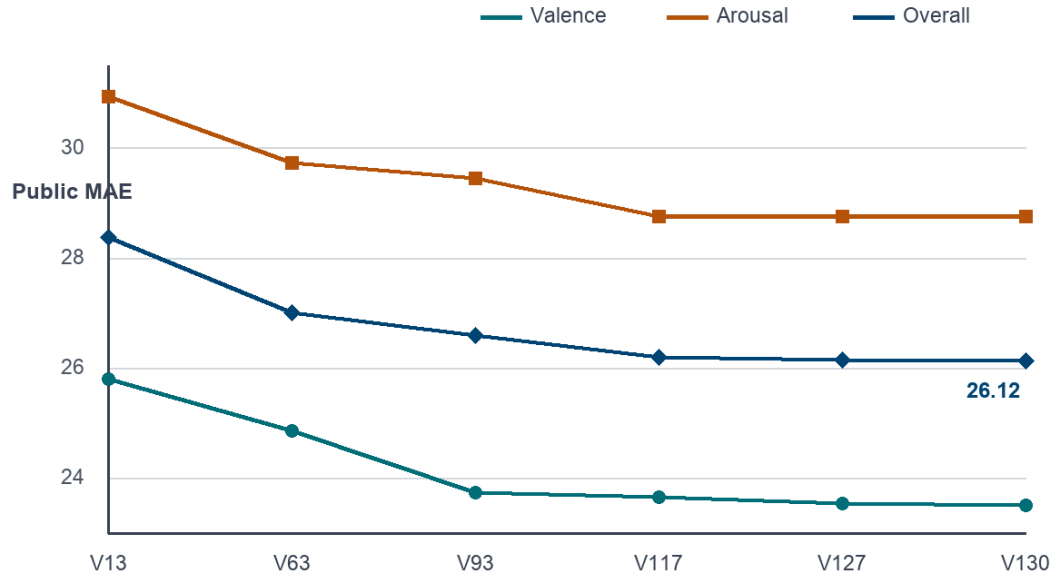
Arousal improved from 30.93 to 28.75 during public development, with most of the gain appearing after chronic rules and agreement-only boosting were introduced. Valence improved from 25.80 to 23.50 as the system moved from global templates to four physiological subtypes and then to the final 11-rule set. These trajectories support the design choice to treat the two affect dimensions differently.

3.3 Transfer behavior of fitted selection

V122 selected from 3,756 candidates by absolute Spearman correlation. It improved local MAE by 0.75 yet worsened the public score by 0.03. V127 accepted a 0.10 local regression in exchange for physiology-guided EEG rules and improved the public score by 0.12; V130 gained another 0.15. The sign reversal is consistent with a few extreme subjects dominating correlation ranks. In a small-cohort setting, weaker but mechanistically constrained effects can be more portable than the largest observed correlations.

Table 2: Final V130 valence rule composition.

Component	Count	Role
Theory EEG rules	11	Stable direction
Theory videos	6	Subtype templates
Fallback features	9	Uncovered videos

**Figure 2: Public-set evolution. Improvements are incremental but consistent; V130 attains 26.12 Overall MAE before final evaluation.****Table 3: Leaderboard evolution and final challenge result.**

Version	V-MAE	A-MAE	Overall	Change
V13	25.80	30.93	28.37	Global median
V63	24.85	29.73	27.01	A rules
V93	23.73	29.44	26.59	Subtype split
V117	23.65	28.75	26.20	A boost
V127	23.53	28.75	26.14	9 EEG rules
V130 pub.	23.50	28.75	26.12	11 EEG rules
V130 final	24.62	26.41	25.52	Rank 7

Table 4: Local improvement did not predict public transfer.

System	Selection	Local dMAE	Public dMAE
V122	3756 x rho	-0.75	+0.03
V127	9 theory EEG	+0.10	-0.12
V130	11 theory EEG	n/a	-0.15

Table 5: Cross-subject outcomes of alternative approaches.

Attempt	Observed outcome
23-subject classifier	+11% train direction, -9% public
IQR-weighted template	MAE 26.97 $\rightarrow$ 29.31
Domain adversarial	Degraded SB2 and Qwen variants
PLV/AEC connectivity	Absolute $r < 0.21$
FAA for valence	High/low accuracy about 50%

3.4 Cross-subject diagnostics

We used three diagnostics to distinguish genuine transfer from memorization. First, feature rankings were recomputed after removing six subjects; a 33% overlap in the top ten indicated high selection variance. Second, direction accuracy was reported separately from MAE. The V14 classifier could raise training direction accuracy by 11% while losing 9% on the public set, showing that better separation of known subjects did not imply better unseen-subject decisions. Third, we inspected output variance. Some spiking runs produced competitive development MAE by converging to an almost constant estimate, which is not evidence of dynamic tracking.

These diagnostics changed the selection objective. A candidate was retained only when its direction was physiologically interpretable, stable across subject subsets, and helpful at the complete-system level. We did not select individual rules directly on final labels. Public feedback was used to decide between named system versions, and the untouched final partition was used for the score in this paper. This procedure cannot remove all leaderboard adaptation, but it prevents thousands of raw candidates from being implicitly optimized against the public score.

4 ANALYSIS

4.1 Valence is physiologically heterogeneous

A single positive-negative axis hides at least four response patterns in our exploratory clustering: approach-positive (V4, V7, V8, V15; frontal asymmetry and gamma increase), immersive-positive (V1, V5, V14; theta and parieto-occipital alpha increase), alert-negative (V3, V9, V11; beta bursts), and avoidance-negative (V2, V6, V10, V12, V13; right-frontal alpha and lower CV). Frontal EEG asymmetry has long been associated with affective style [3], but our results caution against using one global direction for every negative or positive clip. A single hyperplane must reconcile conflicting mechanisms, helping explain the 45.8% trait-direction accuracy for valence.

4.2 Trait dominates state across subjects

The trait/state decomposition is the main empirical explanation for the final system. Resting features primarily identify stable individual baselines, whereas task features primarily identify which clip is being watched. A high-capacity model can exploit both structures on seen subjects, but only the former remains available as a reliable anchor for unseen people. This finding agrees with the broader literature on strong inter-individual variability in EEG emotion recognition [4, 5] and with the need for careful EEG-fNIRS alignment [2].

4.3 Alternative model comparisons

The central limitation was not insufficient model size. Compact regressors reached roughly 40 MAE in early cross-subject tests; spiking models could reach about 24 under development settings but often collapsed toward a near-constant prediction; large-model routes reached about 17 MAE within subject yet remained around 45-50 across subjects. These numbers use different protocols and are therefore diagnostic rather than comparative. Together with Table 5, they show that additional capacity did not solve the subject shift encountered by our pipeline.

4.4 Implications for physiological benchmarks

Sequence length should not be confused with cohort size. Thousands of adjacent seconds can make a training loss numerically stable while leaving the number of independent people unchanged. Evaluation should therefore split by subject before any feature ranking, normalization fit, or hyperparameter choice. Reporting both within-subject and cross-subject scores is also important because our large-model route differed by more than 25 MAE points between those protocols.

The results also argue for dimension-specific baselines. Valence and arousal are usually presented as parallel regression heads, yet they showed different directional reliability and different public-to-final shifts. A competitive benchmark should include robust video priors, per-dimension constant predictors, and trait-only systems in addition to neural baselines. These controls reveal whether a model tracks momentary emotion or mainly estimates a stable subject and stimulus average.

5 DISCUSSION AND LIMITATIONS

Our evidence supports a pragmatic conclusion, not a universal claim that rules always outperform learning. Under the specific MER-PS regime, the effective number of independent subjects is too low to reliably choose among thousands of correlated features. Zero-parameter rules reduce this selection burden and make every decision auditable. They also expose dimension-specific behavior: arousal can use agreement-only amplification, whereas valence requires subtype-aware templates and conservative fallbacks.

Several limitations remain. The four valence subtypes are exploratory and should be validated with larger cohorts. Public leaderboard feedback influenced system-level iteration, so the reported public gains are not an untouched estimate. The final test set provides a stronger check but still comes from the same acquisition protocol. Finally, per-second annotation may be finer than the physiological and behavioral response latency, especially for fNIRS;

future work should evaluate lag-aware and uncertainty-aware targets on external EEG-fNIRS datasets such as REFED [6].

6 REPRODUCIBILITY AND RESPONSIBLE USE

6.1 Reproducibility boundary

The distinction between a zero-parameter inference graph and a zero-decision development process is important. The deployed V130 graph fits no weights, but its rule inventory was chosen through prior experiments on the training and public partitions. To make that boundary explicit, we version every complete system and report its public score in Table 3. The final partition was not used to add, remove, or reweight rules. The score 25.52 therefore evaluates the frozen V130 graph under a new subject split, while the public score 26.12 should be interpreted as development feedback.

A reproduction should preserve three separations. First, subject identities must be split before computing feature ranks or any population statistic. Second, resting-state normalization must use only the current subject’s available baseline and must not borrow held-out annotations. Third, video priors, subtype assignments, confidence thresholds, and fallback mappings must be frozen before final-set prediction. Refitting any of these items on test labels changes the method from a deterministic rule system into a supervised transductive procedure.

The required outputs are equally simple to audit. Every submitted second receives one valence and one arousal value in [1,255]. Overall MAE is the arithmetic mean of the two dimension-specific errors. Results reported on normalized [0,1] labels should be converted before comparison with the challenge leaderboard. We also recommend reporting the standard deviation of predictions by subject and video: a low MAE accompanied by nearly zero output variance may indicate collapse to a constant prior rather than successful trajectory reconstruction.

6.2 Minimum evaluation controls

At least four controls are needed for a fair comparison. A global median estimates how much error can be removed without physiology. A video-specific median tests whether the model mainly recognizes the stimulus. A trait-only system measures the benefit of subject calibration, and a state-only system measures dynamic evidence without that anchor. Our V13, trait/state decomposition, and variance analysis cover these roles. Neural systems should additionally be evaluated with multiple subject splits because one favorable split can mask the instability seen in the 33% feature-overlap test.

Model comparison should keep protocols visible. The within-subject MAE near 17, cross-subject values near 45-50, and public rule score 26.12 answer different questions and cannot be placed in a single ranking without qualification. In particular, random second-level splits allow adjacent windows from the same person and video to appear in both training and validation. Such a split is useful for debugging optimization but overstates deployment to unseen people. Subject-disjoint validation is the primary criterion in this paper.

6.3 Responsible use

Physiological emotion estimates are probabilistic research outputs, not clinical diagnoses or direct readouts of private mental states.

The 1 Hz joystick labels are subjective, and the same signal pattern can reflect motion, attention, fatigue, or individual anatomy. The four valence subtypes should therefore be treated as hypotheses about this benchmark, not fixed categories of people. Applications in health, employment, education, or surveillance would require new consent, external validation, and a clear mechanism for participants to refuse inference.

Releasing only predictions and aggregate errors is not sufficient privacy protection when raw EEG and fNIRS can encode subject identity. Reproducibility packages should follow the dataset license, avoid redistributing identifiable recordings, and remove filenames or metadata that map signals to real people. These constraints are especially relevant to models that learn subject embeddings. Our rule-based design reduces stored model parameters, but it does not eliminate the sensitivity of the underlying physiological data.

7 CONCLUSION

We presented the seventh-place solution to MER2026 Track 4. A zero-parameter ensemble of resting-state trait votes, task-state descriptors, video subtypes, and asymmetric fusion achieved 25.52 final Overall MAE. The strongest lesson is methodological: with 24 training subjects and hundreds of candidate features, local fit is an unreliable proxy for cross-subject transfer. Physiological priors, explicit fallbacks, and dimension-specific rules provided a smaller but more stable hypothesis class.

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